TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC 2 2018

CANDIDATE NAME							
CENTRE NUMBER	S						
INDEX NUMBER							
CLASS	C	G			/	1	7

H1 BIOLOGY

8876/02

Paper 2 Structured Questions

Friday 24 August 2018
2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, Centre number, index number and class in the spaces at the top of the page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graph.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use			
Q1	**/ 6**	**Q5**	**/ 8**
Q2	**/ 10**	**Q6**	**/ 7**
Q3	**/ 5**	**Q7 / 8**	**/ 15**
Q4	**/ 9**		
Total			**/ 60**

This document consists of **16** printed pages.

Answer **all** the questions in this section.

1 Sugar molecules enter cells through transport proteins.

(a) Explain why transport proteins are required for the movement of sugar molecules, such as glucose and fructose, into cells. [2]

1. **Glucose and fructose are polar molecules.**

2. **They are unable to cross**
3. **the hydrophobic core of the phospholipid bilayer.**

4. **Transport proteins shield them from the hydrophobic core of plasma membrane (e.g. channel proteins provide a hydrophilic channel for their movement across the membrane).**

Some plant cells convert fructose and glucose into sucrose for transport from the leaves to the roots. Sucrose is moved into phloem sieve tubes as shown in Fig. 1.1.

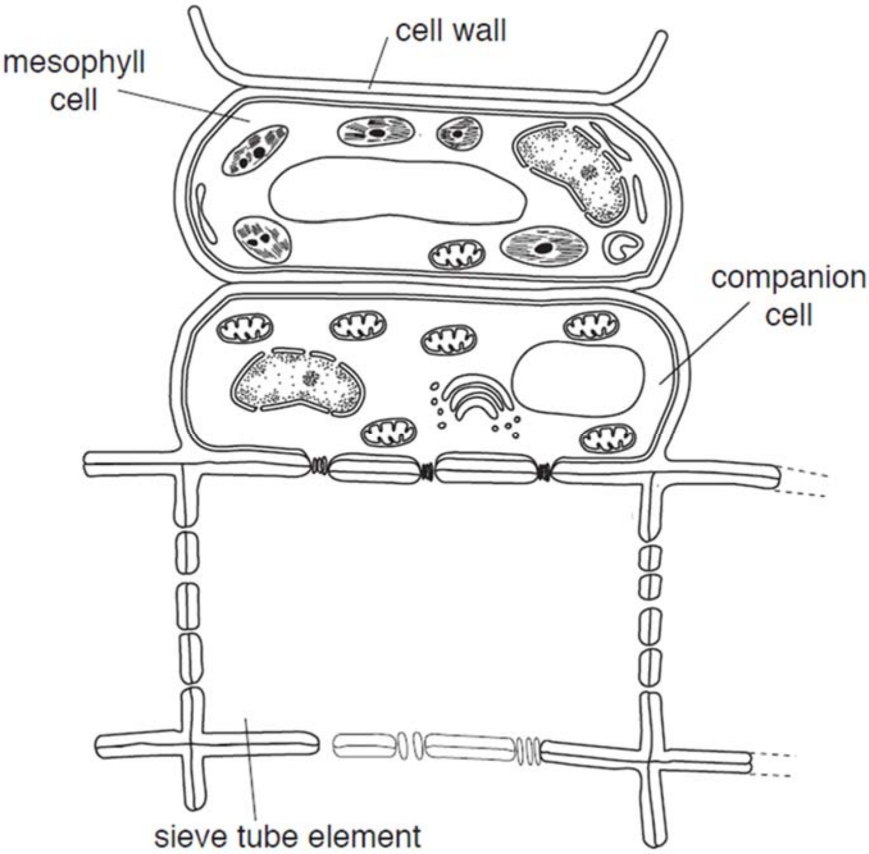

Fig. 1.1

Each cell has a specialized function.

(b) With reference to Fig. 1.1 and the information provided, state **one** difference between a mesophyll cell and companion cell. [1]

1. **Companion cells (6 mitochondria) have more mitochondria than mesophyll cells (1 mitochondrion). [1]**
OR
Mesophyll cells (5 chloroplasts) have chloroplasts whereas companion cells have none. [1]

Fig. 1.2 shows how sucrose is transported into the companion cell from the mesophyll cell.

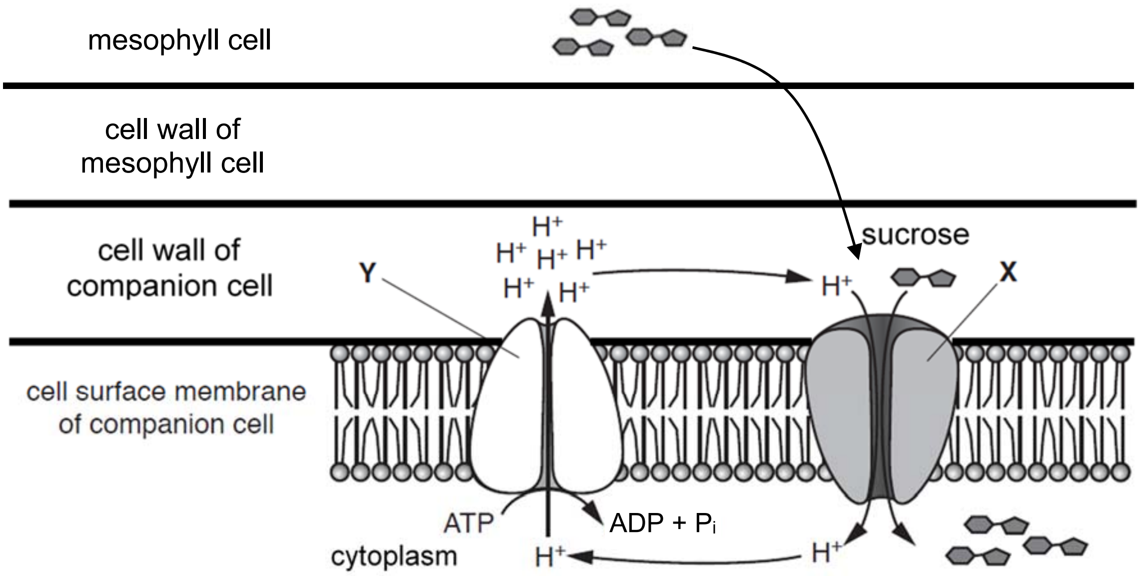

Fig. 1.2

(c) Using the information in Fig. 1.1 and Fig. 1.2, explain how sucrose moves into the companion cell. [3]

1. **Sucrose diffuses from mesophyll cell to the cell wall of companion cell.**

2. **Protons are actively pumped out from the cytoplasm of companion cell into its cell wall through carrier protein Y via active transport (hydrolysis of ATP). [1] [Reject: Diffuse]**

3. **Protons then diffuses from the cell wall of companion cells into the companion cell through transport protein X (cotransporter) via facilitated diffusion [1]**

4. **which is coupled with the transport of sucrose**
5. **against the sucrose concentration gradient.**

[Total: 6]

Extension Question
Plants vary greatly in terms of size.

(d) Explain whether the cell theory is applicable to plants. [2]

1. **Applicable.**
2. **Plants are living organisms, which are composed of (many, different plant) cells,**
3. **which are basic/ smallest unit of life.**

4. **All plant cells come from pre-existing plant cells via cell division (e.g. mitosis or meiosis).**

2 The yeast, *Saccharomyces cerevisiae*, is a single-celled, eukaryotic organism that is often used in the laboratory.

When yeast is mixed with a glucose solution, the yeast absorbs the glucose. Each molecule of glucose is then broken down into pyruvate molecules in exactly the same way as in any other eukaryotic organism.

(a) Outline the breakdown of glucose to pyruvate in this stage. [2]

Respiration Lecture Notes p.9, 10

1. **Glucose is broken to pyruvate during glycolysis.**

2. **Glucose is first phosphorylated to glucose-6-phosphate**
3. **which is (isomerized to fructose-6-phosphate and then phosphorylated to) converted to fructose-1,6-bisphophate**

4. **before being cleaved/ broken down into 2 three-carbon sugars (OR glyceraldehyde-3-phosphate and dihydroxyacetone phosphate),**
5. **which is then oxidised/ converted to form 2 molecules of pyruvate.**

Yeast cells sometimes carry out anaerobic respiration. Fig. 2.1 outlines the process of anaerobic respiration in yeast cells.

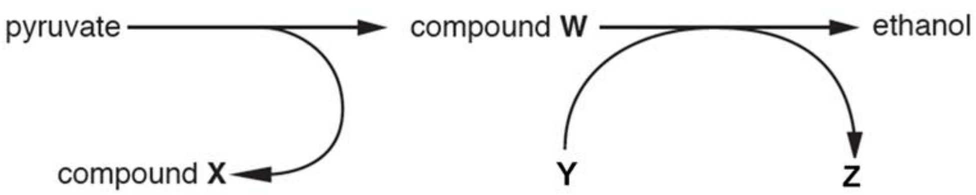

Fig. 2.1

(b) (i) Identify molecule **Z**. [1]

NAD or NAD^+

(ii) State why molecule **Y** is converted to **Z**. [1]

Respiration Lecture Notes p.26

1. **Regenerate NAD**
2. **required for glycolysis to continue.**

[Accept: Reduce compound W (ethanal) to ethanol]

Yeasts are often used in bread-making. The bread dough is kneaded to introduce and trap air so that the yeasts in the dough can respire aerobically. Besides carbon dioxide that is released during respiration, the evaporation of water or ethanol released during respiration also causes the dough to rise.

Table 2.1 shows the differences in the height of dough that was placed at different locations, after the dough was kneaded.

Table 2.1

Time / min	Height of dough / cm		
	Fridge	Room temperature	Next to window (hot day)
0	2.5	2.5	2.5
20	2.5	2.9	3.3
40	2.7	3.7	4.0
60	2.9	3.9	4.7
80	3.0	4.0	5.2
100	3.0	4.0	5.8
120	3.0	4.0	6.0

(c) (i) Account for the difference in the overall increase in the height of dough that was placed in the fridge with that placed next to the window. [4]

1. **The height of the dough when placed in the fridge (F) increases from 2.5 at 0 min to 3.0cm at 120 min**
   is LOWER than
   that when placed next to window (W) which increases from 2.5 to 6.0 cm. [1]

2. **The temperature of the dough in F is lower than that of W.**
3. **Hence, the kinetic energy of respiratory enzymes and substrates is lower.**
   [Accept: Enzymes are inactivated]

4. **The frequency of effective collisions between enzymes and substrates is lower**
5. **hence the rate of formation of enzyme-substrate complexes is lower.**

6. **The rate of respiratory enzyme activity / rate of respiration is lower.**
7. **and less carbon dioxide are released and less evaporation of water or ethanol, which causes the dough to rise less.**

(ii) Suggest why the increase in the height of dough that was placed at room temperature was higher between 0 and 40 minutes than between 40 minutes and 60 minutes. [2]

1. **The height of dough increases from 2.5 to 3.7cm between 0 and 40 min**
   is HIGHER than
   3.7 to 3.9cm between 40 and 60 minutes. [1]

WITH

2. **There are more oxygen between 0 and 40 min, hence the yeast undergoes aerobic respiration which releases more molecules of CO_2 (6 molecules of CO_2**

and 6 molecules of H_2O) than <u>anaerobic respiration</u> (2 molecules of CO_2 and 2 molecules of ethanol) from 40 minutes and 60 minutes. [1]

OR

There are <u>more respiratory substrates</u> at the start between 0 and 40 min, the <u>rate</u> of <u>formation</u> of <u>enzyme-substrate complexes</u> is <u>higher</u>, hence the <u>rate</u> of <u>respiration</u> is <u>higher</u> than between 40 minutes and 60 minutes. [1]

OR

From 40 minutes and 60 minutes, the yeast undergoes <u>anaerobic respiration</u> and <u>high concentration</u> of <u>ethanol</u> produced is <u>toxic</u> and <u>kills</u> the <u>yeast</u>. [1]

[Total: 10]

3 Fig. 3.1 shows the effect of increasing substrate concentration on the rate of a particular reaction in the presence and absence of an enzyme.

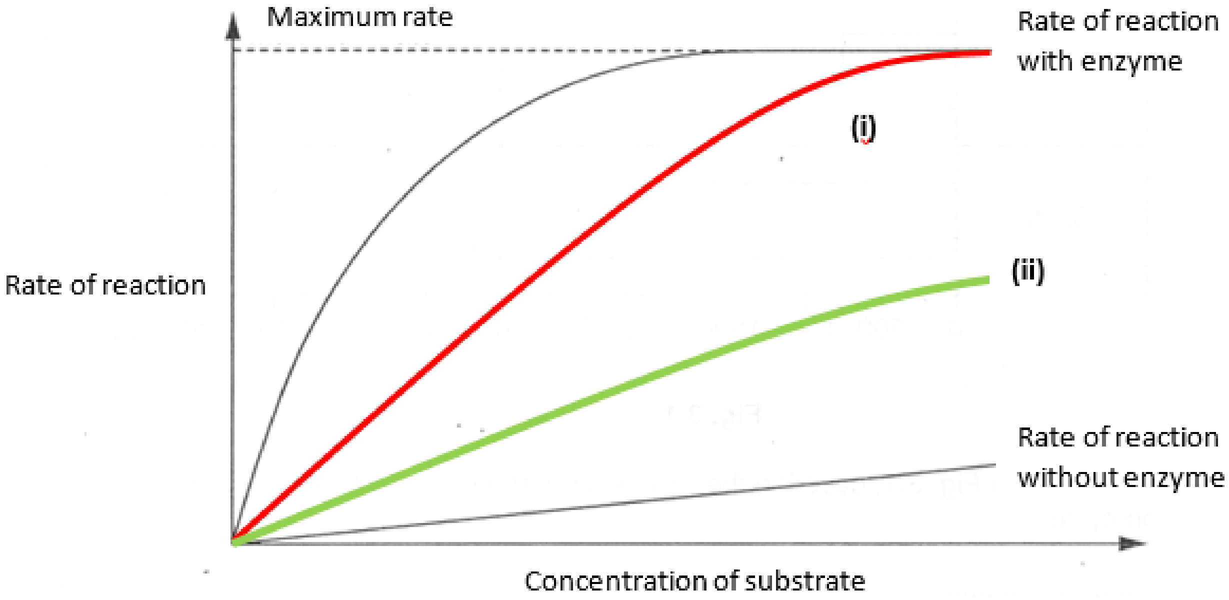

Fig. 3.1

(a) On Fig. 3.1, draw **two** labelled curves to show the effect on the rate of the enzyme catalysed reaction upon the addition of

(i) a competitive inhibitor;

(ii) a non-competitive inhibitor. [2]

(b) Explain the effect of a competitive inhibitor on the rate of enzyme activity. [3]

1. **Shape of inhibitor is similar in shape of substrate**
2. **Shape of inhibitor is complementary to the shape of active site**
3. **Competitive inhibitors compete with the substrate molecules for the active site and bind at the active site of the enzyme**
4. **blocking / prevents substrate molecules from binding to active site,**
5. **reducing**
   - i. **number of enzyme-substrate complex formed per unit time**

     or
   - ii. **rate of enzyme-substrate complex formation**
6. **thus decreasing rate of enzyme activity**

[Total: 5]

4 Table 4.1 shows some of the common fatty acids and their melting points.

Table 4.1

Symbol (number of carbon atoms : number of double bonds)	**Common Name**	**Melting point (°C)**
Saturated fatty acids		
12 : 0	Lauric acid	44.2
14 : 0	Myristic acid	52
16 : 0	Palmitic acid	63.1
18 : 0	Stearic acid	69.6
20 : 0	Arachidic acid	75.4
22 : 0	Behenic acid	81
Unsaturated fatty acids		
16 : 1	Palmitoleic acid	-0.5
18 : 1	Oleic acid	13.4
18 : 2	Linoleic acid	-9
18 : 3	α-linolenic acid	-17
20 : 4	Arachnidonic acid	-49.5

(a) Arachidonic acid is a polyunsaturated fatty acid.

Explain the term *polyunsaturated fatty acid.* [1]

- **A fatty acid with many C=C double bonds.**
  Reject : many kinks

(b) With reference to Table 4.1,

(i) describe the effect of increasing number of carbon atoms in saturated fatty acids on the melting point; [3]

1. As the number of carbon atoms increased 12 to 22, the melting point increased from 44.2 to 81 °C.

2. An initial increase of every 2 carbon atoms from 12 to 18 leads to a sharp increase in the melting point from 44.2 to 69.6 °C.

3. Further increase of every 2 cabon atoms from 18 to 22 lead to a lesser increase in melting point from 69.6 to 81°C.

4. As the number of carbon atoms increases, the melting point increases.

(ii) describe the effect of the presence of double bonds in fatty acids on the melting point; [1]

1. As the number of double bonds increases, the melting point decreases.

2. As the number of double bonds increased from 1 (in oleic acid) to 3 (in α-linolenic acid), the melting point decreased from 13.4 to -17 °C.

(iii) explain the trend described in **b(ii)**. [4]

1. Presence of double bonds results in the fatty acid molecules being bent/ kinked.

2. This means that the molecules cannot be closely packed together / less contact between molecules,

3. resulting in weaker hydrophobic interactions.

4. Therefore, less energy required to overcome the hydrophobic interactions / separate the fatty acid molecules during melting, resulting in the decrease in melting point.

[Total: 9]

5 Table 5.1 provides statements regarding the bonds found in four biological molecules.

Table 5.1

statement	protein	DNA	messenger RNA	cellulose
hydrogen bonds stabilise the molecule	✓	✓	×	✓
subunits are joined by peptide bonds	✓	×	×	×

(a) Complete Table 5.1 by indicating with a tick (✓) or a cross (×) whether the statements apply to proteins, DNA, messenger RNA and cellulose.

You should put a tick or a cross in each box of the table. [2]

(b) A piece of mRNA is 660 nucleotides long but the DNA coding strand from which it was transcribed is 870 nucleotides long.

(i) Explain this difference in number of nucleotides. [1]

- **Introns present in DNA**
- **Introns absent in mRNA**

OR

- **introns removed by RNA splicing**

(ii) What is the maximum number of amino acids in the protein translated from this piece of mRNA? Explain your answer. [2]

Number of amino acids **220 OR 219**
Explanation

1. **3 bases code** for **1 amino acids**

(c) Identify **one** other process that leads to the formation of mature mRNA and state its function. [2]

1. Addition of 5' cap

[Significance]
2. **facilitate the binding of** Translation Initiation Factors and **small ribosomal subunit** for translation to occur.
OR
2. **facilitate the export of mature mRNA** from nucleus to cytoplasm for translation
OR
2. **protect the mature mRNA from degradation** by RNase in the cytoplasm

OR

1. Addition of 3' poly-A tail or 3' polyadenylation

[Significance]
2. **facilitate the export of mature mRNA** from nucleus to cytoplasm for translation
OR
3. **protect the mature mRNA from degradation** by RNase in the cytoplasm

(d) Describe **one** difference between the structure of mRNA and tRNA. [1]
Any one:
1. **mRNA** has **no base-pairing within** its **structure** while **tRNA** has **base-pairing between** regions to **fold back on itself**.
2. **mRNA** has **3' poly-A tail** while **tRNA** has **3' CCA end**.
3. **mRNA** does **not** have **hydrogen bonds** different regions of the single strand while **tRNA** has **hydrogen bonds** at different regions which cause it to **fold back on itself**.
4. **mRNA** is **linear** while **tRNA** **cloverleaf** shape;
5. **mRNA** has **no binding site for amino acids** while tRNA has.
6. **mRNA** **longer**/larger/more nucleotides than tRNA
7. **Mrna** **different** for **each gene**/many kinds, **only few/**20/64 **kinds of tRNA**;

6 The evolutionary origin of the four-legged amphibians (such as frogs and toads) from fish has been the subject of much debate for many years.

Among living fish, the rarely-caught coelacanth and the lungfish are thought to be most closely related to these amphibians.

Samples of blood were taken from two coelacanths that were captured recently near Comoros.

The amino acid sequences of the α and β chains of coelacanth and lungfish haemoglobin were compared with the known sequences of amphibian adults and their aquatic larvae (tadpoles).

Organisms with more matches in the amino acid sequence of a polypeptide chain share a more recent common ancestor than those with fewer matches.

The comparisons with three species of amphibians, *Xenopus laevis* (Xl), *X. tropicana* (Xt) and *Rana catesbeiana* (Rc) are shown in Table 6.1.

Table 6.1

		percentage of matches of amino acid sequence					
		species of amphibian adults			species of amphibian larvae (tadpoles)		
	fish species	*Xl*	*Xt*	*Rc*	*Xl*	*Xt*	*Rc*
α chains	coelacanth	42.0	47.5	no data	45.4	42.6	48.2
	lungfish	40.4	42.1	no data	40.7	39.0	37.9
β chains	coelacanth	42.1	43.2	40.7	52.1	52.1	58.2
	lungfish	44.1	45.9	41.4	47.3	45.9	48.6

(a) Explain whether or not the information in Table 6.1 supports the suggestion that coelacanths and amphibians share a more recent common ancestor than do lungfish and amphibians. [4]

1. **Data largely support that coelacanth and amphibians share a more recent common ancestor. [1]**

2. **The α chain of all 3 species of ADULT amphibians have a higher match with that of coelacanth (42 and 47.5) than lungfish (40.4 and 42.1). [1]**

3. **The <u>α chain</u> of <u>all</u> 3 species of <u>LARVAL amphibians</u> also have a <u>higher</u> match with that of <u>coelacanth</u> (<u>45.4</u>, <u>42.6</u> and <u>48.2</u>) than <u>lungfish</u> (<u>40.7</u>, <u>39.0</u> and <u>37.9</u>). [1]**

4. **Only the <u>β chain</u> of <u>all</u> 3 species of <u>LARVAL amphibians</u> (rather than adults) have a <u>higher</u> match with that of <u>coelacanth</u> (<u>52.1</u>, <u>52.1</u> and <u>58.2</u>) than <u>lungfish</u> (<u>47.3</u>, <u>45.9</u> and <u>48.6</u>). [1]**

(b) Coelacanth haemoglobin has a very high affinity for oxygen, suggesting that coelacanths, which have been captured at depths of between 200 m and 400 m, live in water that has a low concentration of oxygen.

Explain how an environmental factor, such as the low concentration of oxygen in deep water, can act as an evolutionary force in natural selection. [3]

1. **<u>Low oxygen</u> concentration acts as <u>selection pressure</u>.**

2. **<u>Individuals</u> with <u>haemoglobin</u> with a <u>higher affinity</u> to <u>oxygen</u> are <u>better adapted</u> to <u>low oxygen concentration</u> are at <u>selective advantage</u>.**

3. **They <u>survive</u> to <u>reproductive age</u> to <u>produce</u> <u>viable</u> and <u>fertile</u> <u>offspring</u>,**
4. **hence <u>passing</u> their <u>favourable alleles</u> to their offspring.**

5. **This leads to an <u>increase</u> in <u>frequency</u> of <u>favourable alleles</u> in population, leading to <u>more individuals</u> with <u>adaptation</u> for <u>low oxygen concentration</u>.**

6. **<u>Directional selection</u> occurred.**

[Total: 7]

✄ End of Section A ✄

2018 PRELIMINARY EXAMINATION

H1 BIOLOGY PAPER 2 [SECTION B]:

Essay Question

Name: ________________________________

Civics Group: _________/17

For Examiner's Use	
Q7 / 8	**/ 15**

Section B

Answer **one** question.
Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must set out in sections **(a)**, **(b)** etc., as indicated in the question.

EITHER

7 (a) Discuss the suggestion that all living organisms on earth depend on nitrogen. [6]

1 mark EACH

[Proteins/ Enzymes] **[Prot]**

1. **Amino group (containing N) of amino acid → Form peptide bond to form polypeptides / proteins → metabolic processes (e.g. enzymes for respiration)**

[Haemoglobin] **[Hb]**

2. **Nitrogen in porphyrin ring of haem in haemoglobin → Take up and release oxygen**

[Phospholipid] **[PL]**

3. **Nitrogen in choline bonded to phosphate head of phospholipid → Maintain cell structure / facilitate cell signaling / transport across membrane**

[Nucleotides and nucleic acids] **[NRep]**

4. **Nitrogenous bases in DNA (e.g. Adenine, Guanine, Cytosine, Thymine) → Stability of DNA structure via complementary base pairing**

5. **Nitrogenous bases in DNA (e.g. Adenine, Guanine, Cytosine, Thymine) → Hereditary material that is passed on to the offspring / Act as template for synthesis of daughter strands via complementary base pairing**

[Nucleotides and nucleic acids] **[DNA/ RNA-Transc/ RNA-Transl]**

6. **Nitrogenous bases in DNA (e.g. Adenine, Guanine, Cytosine, Thymine) → Store genetic information / Act as template for the synthesis of mRNA via complementary base pairing during transcription.**
   OR
7. **Nitrogenous bases in RNA (e.g. Adenine, Guanine, Cytosine, Uracil) → Act as template → Form complementary base pairing between codon of mRNA and anticodon of tRNA during translation.**

[Respiration / Cellular activities: ATP] **[ATP]**

8. Nitrogen is found in structure of ATP → Hydrolysis of high-energy bonds of ATP to provide energy for cellular/ metabolic activities
   [Reject: Produce energy]

   [Accept: Other roles of ATP: Phosphorylation of glucose / fructose-6-phosphate for glycolysis; Involvement in Calvin cycle; Amino acid activation]

[Respiration: NAD/ FAD] [NAD/ FAD]

9. Nitrogen is found in structure of NAD → Electron carrier in glycolysis, link reaction and Krebs cycle → Donate electrons and protons for oxidative phosphorylation to synthesize ATP

10. Nitrogen is found in structure of FAD → Electron carrier in Krebs cycle → Donate electrons and protons for oxidative phosphorylation to synthesize ATP.

[Respiration: $NADP^+$] [$NADP^+$]

11. Nitrogen is found in structure of $NADP^+$ → Final electron acceptor of non-cyclic photophosphorylation to form NADPH → Reduce glycerate-3-phosphate to glyceraldehyde-3-phosphate in Calvin cycle

[Photosynthesis: Chlorophyll] [Chl]

12. Nitrogen is found in structure of chlorophyll → Photosynthetic pigments that harvest light energy during photophosphorylation to produce ATP and NADPH for Calvin cycle to occur.

[Respiration: GTP] [GTP]

13. Nitrogen is found in structure of GTP → Product of substrate-level phosphorylation in Krebs cycle before conversion to form ATP

    [Accept: Other roles of GTP: Translocation of ribosomes during translation]

[QWC] [QWC]

14. Paragraphing + At least 2 different biomolecules mentioned

(b) Discuss the extent to which mitosis and the different types of stem cells can account for the principles behind the cell theory in humans, from one generation to the next. [9]

[Cell Theory] [CT]

1. All living organisms are made up of cells.
2. Cell forms basic unit of life.
3. All cells come from pre-existing cells via mitosis.

4. New cells arise from stem cells.

[From parental cell to daughter cells within individual and description of stem cells in terms of contribution to cell theory]

[F]

5. In humans, life started from the fusion of gametes to form a zygote during fertilisation. [1]

[TSC]

6. Zygotic stem cells are totipotent which can give rise to all cell types. [1]

[PSC]

7. Embryonic stem cells are pluripotent stem cells give rise to almost all cell types. Inner cell mass (ICM) forms all different cells and tissues of human body. [1]

[MSC]
8. Multipotent stem cells (e.g. blood stem cells) give rise to a limited range of cell types. [1]

[From one parental generation of organism to the next generation] [M]
9. From one generation to the next, gametes are produced by meiosis. [1]

[Comment on extent] [C]
10. Stem cells and mitosis account for cell theory within organisms but not from one generation to the next (meiosis and fertilization). [1]

[QWC] [QWC]
11. Paragraphing + Explains cell theory + Explain potency of stem cells + Comment [1]

OR

8 (a) Discuss possible impact of global warming on geographical patterns of distribution of mosquito-borne diseases. [6]

1 mark EACH:
[Example of disease] [E]
1. Increase in mosquito-borne diseases (e.g. dengue) in Singapore.

[Geographic distribution: Range] [R]
2. Global warming will result in increase in range in terms of latitude (geographical location), e.g. spread from equator to subtropical regions (e.g. Europe)

[Geographic distribution: Altitude] [A]
3. and altitude (elevation) from plains to hills (e.g. Nepal)

[Impact: Increased temperature on mosquitoes' survival] [MS, DT]
4. Increased temperature (up till 32°C) increases the survival and reproduction of the mosquito,
5. and the female mosquitoes bite more often, increasing the transmission of dengue.

[Impact: Increased temperature on viral replication] [VR]
6. Increased temperature may negatively affect the reproductive cycle of dengue virus, as viral enzymes may be denatured, hence reducing the number of dengue virus and decreasing the transmission of dengue.

[Impact: Increased rainfall on breeding grounds] [BG]
7. Increased rainfall may result in more stagnant water and increases the number of breeding habitats for mosquitoes.

[QWC] [QWC]
8. Paragraphing + Explains geographical changes based on reasoning linked to 2 aspects of climate change (temperature and rainfall) [1]

[Accept: Other trends with justified claims]

(b) DNA molecules are replicated with a high degree of accuracy yet not always perfectly. [9]

Describe how this occurs and discuss why the survival of a species depends on DNA molecules being stable, yet not *absolutely* stable.

1 mark EACH:

[High degree of accuracy: Complementary base pairing] [SCR, CBP]

1. **During semi-conservative DNA replication, each parental strand acts as a template for the synthesis of daughter strand.**

2. **Complementary base pairing occurs between nucleotides of template strand and free nucleotides.**

3. **DNA polymerase proofreads the newly synthesized daughter strand (replace incorrectly paired deoxyribonucleotides) / DNA repair occurs to replace damaged DNA strands/ OWTTE.**

[High degree of accuracy: Importance of genetic stability] [GI]

4. **This ensures that daughter cells (NOT strands) have genetically identical DNA as parental cell for growth and development of a multicellular organism/ replacement of worn-out parts of the body/ asexual reproduction.**

[Imperfect accuracy: Mutation] [M, ME]

5. **Errors in proofreading/ DNA repair mechanisms can result in mutation,**
6. **(e.g. insertion, deletion, substitution) whereby sequence of nucleotides in a gene is changed.**

[Imperfect accuracy: Disease] [D]

7. **This may result in diseases (e.g. cancer), or genetic disease that threatens the survival of the organism.**
   E.g. Sickle cell anaemia caused by a single nucleotide substitution (T→A), which results in mutant haemoglobin and sickle-shaped red blood cells that results in the less efficient transport of oxygen which may reduce the survivability of the individual.

[Imperfect accuracy: Genetic variation] [NA]

8. **Gene mutation results in the formation of new alleles / is the ONLY source of new alleles, thus increasing the gene pool (genetic variation) and results in new phenotypes.**

[Imperfect accuracy: Impacts of genetic variation on natural selection] [E]

9. **Genetic variation (diversity) within a population is crucial to the survival of species especially when there is a change in the environment/ selection pressure/ allows them to best adapt to the environment.**
   OR
   Individuals that have the favourable alleles are at selective advantage and they are selected for, thus they are more likely to survive and reproduce to produce more viable and fertile offspring, thus passing their favourable alleles to their offspring.

[Conclusion] [C]

10. **Hence, the continuity of a species and its continued evolution relies on a balance between accurate transmission of nucleotide sequences to the offspring and variation needed to allow continued evolution of the species such that it does not lead to death, thus allowing the species to respond to environmental changes/ OWTTE.**

[QWC] [QWC]

11. **Paragraphing + Covers all requirements of questions (i.e. describe and explain stability and instability of DNA) + Explain how it affects survival of species**